EXERCISE 3

Ordinary Differential Equations

universität wien
Faculty of Physics

We study the chemical reaction

$$\mathrm{A} \xrightarrow{k_1} \mathrm{B} \xrightarrow{k_2} \mathrm{C}$$

where A, B, C are the concentrations of three substances and k_1, k_2 the reaction rates. Initial values $A(0) = 1$, $B(0) = C(0) = 0$, rates $k_1 = 1$ and $k_2 = 1, 10, 100, 1000$, time $t \in [0, 10]$. Analytical solution for C:

$$C(t) = 1 - \frac{k_2\, e^{-k_1 t} - k_1\, e^{-k_2 t}}{k_2 - k_1} \quad (k_1 \neq k_2), \qquad C(t) = 1 - e^{-k_1 t}(1 + k_1 t) \quad (k_1 = k_2)$$

1 Chemical reaction.

a) Derive the standard form $\dot{\mathbf{x}} = J\mathbf{x}$ of the problem, $\mathbf{x} = (A, B, C)$.
Why does $A + B + C$ stay constant?

b) Implement explicit Euler, Heun, RK4 and implicit Euler (for implicit Euler solve $(\mathbf{1} - \Delta t\, J)\,\mathbf{x}_{n+1} = \mathbf{x}_n$)
How can you check that the results are correct?

c) Compare the accuracy of the methods for varying Δt ($k_2 = 10$).
Which slopes do you expect in a log-log plot of the error, which do you get?

d) Check for which Δt the methods become unstable, and how the critical time step depends on k_1 and k_2.
What does the solution look like just above the critical time step?

e) Calculate the eigenvalues λ and eigenvectors of J. Visualize the stability regions $|R(z)| \leq 1$ of the methods together with $z = \lambda \Delta t$ for different Δt, and the eigenmodes.
Which eigenmode becomes unstable first? Why is implicit Euler always stable?

f) **Stiffness** ($k_2 = 1000$). How many steps do explicit and implicit Euler need for an error of about 10^{-2}?
What limits Δt in each case?

2 More complicated equations.

Apply your methods to the following nonlinear ODEs and try to find interesting solutions.
Which method do you trust, and how do you check it without an exact solution?

a) Robertson model (stiff chemistry):

$$\dot{A} = -k_1 A + k_3 BC$$
$$\dot{B} = k_1 A - k_2 B^2 - k_3 BC$$
$$\dot{C} = k_2 B^2$$

Solve e.g. for $k_1 = 0.04$, $k_2 = 3 \cdot 10^7$, $k_3 = 10^4$, $(A, B, C)(0) = (1, 0, 0)$, $t \in [0, 10^5]$ (plot over $\log t$). Implicit Euler: solve $\mathbf{x}_{n+1} - \mathbf{x}_n - \Delta t\, \mathbf{f}(\mathbf{x}_{n+1}) = 0$ with Newton's method.
Why do explicit methods fail here?
Could you directly reuse your code from 1?

b) Lorenz system (chaos):

$$\dot{x} = \sigma\,(y - x)$$
$$\dot{y} = x\,(\rho - z) - y$$
$$\dot{z} = xy - \beta z$$

Solve e.g. for $\sigma = 10$, $\rho = 28$, $\beta = \frac{8}{3}$, $t \in [0, 40]$, from $(1, 1, 1)$ and from $(1, 1, 1 + 10^{-8})$.
When do the two trajectories separate?